

Nicola Saltarelli

 LinkedIn |  GitHub |  Foggia, Italy

EDUCATION

Politecnico di Bari

Master of Science in Automation Engineering

Oct. 2023 – Oct. 2025

Bari, Italy

- **Final grade:** 110/110 with honors | **GPA:** 3.92/4.0

Politecnico di Bari

Bachelor of Science in Computer and Automation Engineering

Oct. 2020 – Jul. 2023

Bari, Italy

- **Final grade:** 110/110 with honors

EXPERIENCE

Politecnico di Bari

Graduate Researcher

Nov. 2025 – Present

Bari, Italy

- Conducting advanced R&D in Robotics, AI, and Computer Vision, translating theoretical research into scalable algorithms for industrial applications.
- Collaborating with industrial partners to solve complex operational challenges through tailored automation and artificial intelligence solutions.
- Co-authoring peer-reviewed papers presented at international conferences.

PUBLICATIONS

Online Reinforcement Learning for Adaptive Gain Tuning in Image-Based Visual Servoing

2026

- N. Saltarelli and P. Lino. In Proc. of the 34th Mediterranean Conference on Control and Automation (MED), Ancona, Italy. [Presented, available on IEEE Xplore]

A Hierarchical A* - SAC Architecture for Safe and Efficient AUV Path Planning

2026

- N. Saltarelli, M. Cosmo and P. Lino. Submitted to the International Conference on Mechatronics - Mechatronika (ME), Prague, Czech Republic. [Under Review]

PROJECTS

Deep Learning Anomaly Detection for Industrial IoT (*Industrial Project*) | Python, TensorFlow

- Developing a predictive maintenance framework leveraging Deep Autoencoders to detect anomalies in time-series telemetry data from industrial equipment.

Robust Control for Linear Actuators (*Industrial Project*) | MATLAB, Simulink

- Engineered an automated H_∞ controller synthesis procedure optimized via Particle Swarm Optimization (PSO) for industrial linear actuators. This approach drastically reduced position overshoot to prevent component breakage, boosting overall production throughput by over 20%.

LPBF thermal monitoring deep learning (*Industrial Project*) | Python, MATLAB

- Engineered a Computer Vision pipeline based on a modified ResNet-18 architecture to monitor Laser Powder Bed Fusion (LPBF) processes, classifying laser power and scanning speed from thermal images.

Turtlebot3 Autonomous Exploration | ROS 2, Gazebo

- Developed and simulated an autonomous exploration system for a TurtleBot3 robot leveraging ROS 2 and Gazebo.

Motion Control with Neural Network | Simulink, MATLAB

- Implemented a two-layer neural network for the motion control of a 6-DOF robotic manipulator (Puma 560), utilizing Simulink and Peter Corkes Robotics Toolbox.

SKILLS

Coding: Python, C++, Java, MATLAB, R, SQL, JavaScript, HTML, CSS, NodeJS, ReactJS, VHDL

Frameworks & Tools: ROS2, Nav2, SLAM, Docker, PyTorch, TensorFlow, Scikit-learn, Stable-Baselines3, OpenCV, MoveIt, Gazebo, Simulink, Solidworks, Labview

Languages: Italian (Native), English (B2)